LESSON 5.4
MAKING POPULATION PREDICTIONS – PART 2

LESSON INTRODUCTION

MA.7.DP.1.3 Given categorical data from a random sample, use proportional relationships to make predictions about a population.

LEARNING TARGETS

- I can use proportional relationships to make predictions about a population.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey and develop a margin of error through the use of simulation.

ESSENTIAL QUESTIONS

- How can data from a sample be used to make predictions about a population?
- How can fractions and percentages be used to make population predictions?

LESSON NARRATIVE

Students continue their exploration of using proportional relationships to make predictions about populations from sample data. Students complete a formative assessment as Bell Work, then work through analyzing strategies and interpreting solutions under the teacher's guidance. Students then synthesize all they've learned to complete a Collaboration where they discuss and choose the best strategies to determine how to stock a school vending machine based on sample data. After independent practice, students complete a formative assessment as a check for understanding of the learning target.

COMPONENT SUMMARY

5.4.1 WARM-UP *(~5 MIN.)*

Predict the number of fish that will be caught based on previous data

5.4.2 GUIDED PRACTICE *(10-15 MIN.)*

Discuss and analyze strategies used during Guided Practice

5.4.3 COLLABORATION *(10-15 MIN.)*

Collaborate to make predictions about how to stock a vending machine based on student survey data

5.4.4 YOUR TURN! *(10 MIN.)*

Make predictions about a population based on sample data

5.4.5 WRAP-UP *(~5 MIN.)*

Formative assessment of the lesson's learning target

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RESOURCES

GLOSSARY

Key Terms:

- Scale factor
- Equivalent fraction
- Percent

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble choosing the strategy that is best for each situation or struggle to reason about which is the most efficient.

THINGS TO CONSIDER

- At this time, the focus in problem-solving for these types of questions is NOT on using proportions. Unit 6 will focus on setting up and solving proportions. The focus in this unit is on using strategies such as equivalent fractions, scale factor, and percentages.
- In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. If not given a place value for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

5.4.2 Guided Practice positions students to consider and reason about which strategy makes the most sense for them to use and why. Encourage students to build on their partner's responses or make changes to theirs if they find that one strategy may be more beneficial than their original.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

5.4.3 Collaboration provides students an opportunity to work together and present a variety of ideas for how to use the sample data to solve a problem. Consider providing opportunities for students to present their strategies and ideas to the class so that they can have the opportunity to analyze the mathematical thinking of others. For example, class time could be spent here and the 5.4.4 Your Turn! could be assigned for homework instead.

DIFFERENTIATE INSTRUCTION

Some students may be overwhelmed by choosing from the three strategies, or it may be the case that only certain strategies make sense to them. Students can be given two choices instead of three. For those students who are overwhelmed by having to choose from even two strategies, allow them to use one strategy. Students will continue working with proportional reasoning and these strategies throughout the upcoming units, so it is not necessary that they understand all strategies well at this point in instruction.

Additional differentiation opportunities are denoted throughout the lesson guide.

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5.4.1 WARM-UP: BELL WORK

TEACHER MOVES

Consider positioning students to discuss their strategies for determining the number of fish to be caught on just one Sunday. This gives students an opening to what they'll be working on in this lesson and connects to the idea of reasonable versus unreasonable predictions from the prior lesson.

5.4.1 Warm-Up: Bell Work

Antonia and her aunt go fishing at Lake Okeechobee every Sunday. For the past four Sundays, Antonia has kept track of what kinds of fish they caught.

Type of Fish	Bluegill	Bream	Crappie	Largemouth Bass
Number of Fish Caught Last Month	10	12	20	8
Expected Number of Fish to Be Caught Next Sunday	3	3	5	2

1. Complete the table to show what type and how many fish Antonia and her aunt are likely to catch next Sunday when they go fishing.

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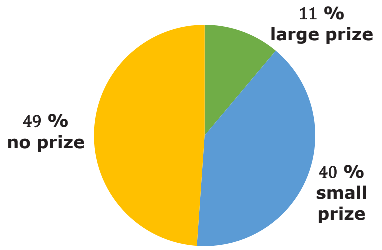
5.4.2 Guided Practice: Making Population Predictions

1. Ella gathered data from a Rubber Duck Pool Game. The circle graph below shows the percentage of people who won a large prize, a small prize, and no prize.

Prizes Won in Rubber Duck Game



There are 180 rubber ducks floating in the pool. The person playing the game takes out one duck and looks for a mark on the bottom of it.

- If there is a blue mark, the person wins a small prize.
- If there is a green mark, the person wins a large prize.
- If there is no mark, the person does not win a prize.

Ella uses her data to predict how many of each kind of duck are in the pool using three different strategies. Her work is shown.

Method 1	Method 2	Method 3
40% of all ducks	11% of all ducks	49% of all ducks
$\frac{40}{100}=\frac{4}{10}=\frac{72}{180}$	$0.11 \times 180 = 19.8$	$\frac{49}{100} \times \frac{0.56}{0.56} = \frac{27.44}{180}$
So, 72 ducks have a blue mark.	So, 20 ducks have a green mark.	So, 27 ducks have no mark.
Equivalent fraction	*Percent*	*Scale factor*

a. Ella's solution ○ is ○ is not correct because

○ the number of ducks with green marks is less than the number of ducks with blue or no marks.
○ the number of ducks with no marks is less than the number of ducks with blue marks.
○ Ella's solution shows a total of 180 ducks.

b. Write an explanation for why you agree or disagree with Ella's solution. Use specific examples from Ella's strategies to justify your opinion.
I do not agree with Ella's solution. Her answer does not total to 180. If you add the terms together, you have 119 ducks. It looks like her mistake was in the third method. She should have multiplied the numerator and denominator by a scale factor of 1.8, not 0.56. This would have resulted in 88.2, or 88 ducks.

c. Compare your explanation with your partner's. Discuss any differences and make changes if necessary. *Answers will vary.*

d. Ella used three different solution strategies. In the bottom of each column in Ella's table, label each strategy with the correct name from the descriptions.

5.4.2 GUIDED PRACTICE: MAKING POPULATION PREDICTIONS

TEACHER MOVES

Consider prompting students to mark the text using the Mark-Up strategy as they read and understand the context. `

ENRICHMENT

Students may struggle to determine whether or not Ella's solution is correct. Encourage students to consider the following questions to reason about Ella's response and continue to use the pie chart as a reference:

- What do Ella's answers add up to? What should the total be?
- Even before completing the calculations, what group should the largest number of ducks be? How do you know? The smallest amount? Do these match Ella's answers?

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5.4.2 GUIDED PRACTICE: MAKING POPULATION PREDICTIONS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Labeling each of Ella's strategies provides a reference point for students to return to as they work to solve similar problems on their own.

TEACHER MOVES

Think-Pair-Share

Position students to discuss which strategy they think will work best and why before completing question 2b in writing. This will allow students to reason about each of the strategies and potentially hear why another strategy might be more efficient or make more sense. Students have the opportunity to solidify their thinking before putting it in writing.

When using a **scale factor,** first divide the population by the sample to find the scale factor, then multiply the numerator and denominator of the sample fraction by the scale factor.

When using an **equivalent fraction,** write multiple fractions equal to the sample fraction until the denominator shows the total population.

When using a **percent,** multiply the total population by the decimal or fraction form of the percent of the sample.

2. Bianca picks blocks out of a bag one at a time. She picks out a block, records the color, and puts the block back in the bag. She cannot see the blocks, but she knows there are 6 blocks in the bag. Of the 50 times she picked a block out of the bag, she picked a blue block 29 times, a red block 10 times, and a yellow block 11 times.

   a. Write ratios to show the number of times Bianca picked a blue, red, and yellow block out of the bag.

   Blue $\frac{29}{50}$ Red $\frac{10}{50} = \frac{1}{5}$ Yellow $\frac{11}{50}$

   b. Explain which strategy will work best to predict how many of the 6 blocks are blue, red, and yellow.
   Answers will vary.
   Sample answer: The strategy of percent would be the best strategy to use in this case since we already have the fraction form of the percent from the sample from part a.

   c. Using the strategy you selected, how many of the 6 blocks are blue, red, and yellow? Show your work.
   Answers will vary.

   Percent Strategy:
   Blue: $\frac{29}{50} \times 6 = 3.48 \approx 4$
   Red: $\frac{10}{50} \times 6 = 1.2 \approx 1$
   Yellow: $\frac{11}{50} \times 6 = 1.32 \approx 1$

   Therefore, there is 1 red block, 1 yellow block and 4 blue blocks.

   d. Explain whether the strategy you chose was the best strategy to use, or if another would have worked better.
   Answers will vary. The strategy that I chose was the best because we already knew the fraction form of the percent from the sample in part a. Using equivalent fractions could be tricky because the least common multiple for 6 and 50 is 300. Using scale factor would require additional work to first determine that the scale factor is $\frac{6}{50} = 0.12$, and then multiple the scale factor by numerator and denominator of the sample fraction.

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5.4.3 Collaboration: Stocking a Vending Machine

Students at Sanford Middle School are upset the school vending machine is always sold out of their favorite items. The student council decides to survey students to find a better way to stock the vending machine. They randomly sample 60 students. Each student surveyed picked their 5 favorite items from the vending machine. The results are shown in the table.

Item	Number of Times Chosen in Survey	Number that Should be Stocked
Reese's	30	45
Skittles	24	30
Starburst	21	15
Snickers	42	75
Pretzels	24	30
Chips	30	45
Takis	39	75
Gum	36	60
M&Ms	30	45
Trail Mix	24	30

The vending machine at school has 30 different snack compartments, each with 15 metal spirals for each snack. The vending machine can hold 450 total snacks.

1. Work with your partner to decide how many of each snack should be stocked in the vending machine. Complete the table with your decision.

   Your work will be assessed using these criteria.
   - Each of the strategies: scale factor, equivalent fraction, and percent are used. If one of the strategies is not used, there is an explanation for why the strategy was not used.
   - It is clear, either from your work or a written explanation, why you have stocked the vending machine the way you did.
   - The number of each snack stocked in the vending machine is a direct reflection of the survey results.

Item	Number of Times Chosen in Survey	% of Students Who Selected Item in Top Five Favorite	# of Rows Each Item Should Have	Number that Should be Stocked
Reese's	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Skittles	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$
Starburst	21	$\frac{21}{60} = \frac{7}{20} = 35\%$	1	$1 \times 15 = 15$
Snickers	42	$\frac{42}{60} = \frac{21}{30} = 70\%$	5	$5 \times 15 = 75$
Pretzels	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$
Chips	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Takis	39	$\frac{39}{60} = \frac{13}{20} = 65\%$	5	$5 \times 15 = 75$
Gum	36	$\frac{36}{60} = \frac{3}{5} = 60\%$	4	$4 \times 15 = 60$
M&Ms	30	$\frac{30}{60} = \frac{1}{2} = 50\%$	3	$3 \times 15 = 45$
Trail Mix	24	$\frac{24}{60} = \frac{2}{5} = 40\%$	2	$2 \times 15 = 30$

Use the survey results to find the percentage of surveyed students who selected the snack in their top five favorite. Based on preference, determine how many rows each item should have in the vending machine. In the table above, the rows were determined by the following:

$0\% - 35\% = 1$ *row*
$36 - 45\% = 2$ *rows*
$46\% - 55\% = 3$ *rows*
$56\% - 65\% = 4$ *rows*
$66\% - 100\% = 5$ *rows*

Since we know there are 15 snacks in each row of the vending machine, use the scale factor of 15 to determine to total number of each item.

5.4.3 COLLABORATION: STOCKING A VENDING MACHINE

TEACHER MOVES

Students may have trouble choosing the strategy that is best for each situation. Since all of the strategies work in all situations, they have flexibility in their choice.

TEACHER MOVES

Consider the different ways students might approach this problem and circulate as they work to identify unique strategies. There is not one correct answer to this problem.

ENRICHMENT

Further scaffold this section for students as they work if needed. Consider posting these questions somewhere for students to reference or circulating and positioning students to consider and discuss as you see fit:

- What do you notice about the size of the sample compared to the size of the population?
- How many times larger/smaller is the population than the sample?
- What number can be multiplied by the sample size to get the total in the population?
- If the gum was chosen 42 out of 300 times, what percentage is that? (This means gum should make up 14% of the items in the vending machine.)
- How would we determine how many spots out of 450 the gum would take up? ($\frac{42}{300} = \frac{21}{150} = \frac{63}{450}$ of the items in the vending machine.)

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5.4.4 YOUR TURN!

TEACHER MOVES

Students may struggle to choose a strategy to predict the number of students who take the bus or ride their bikes to school. Remind students of the three strategies they have been practicing throughout this lesson and encourage them to use the strategy that makes the most sense to them.

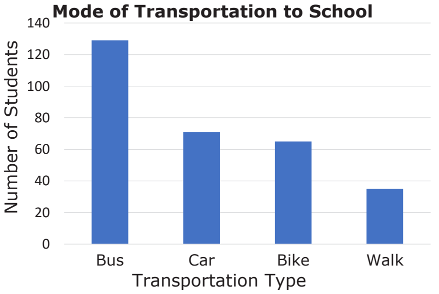
ENRICHMENT

If students are struggling to answer question 1c, consider posing the following questions:

- Does this data set show how far students travel to get to school?
- Why might students ride their bikes? Is it possible that students who ride their bikes live in the area where they could take the bus?
- Why might the district want to know how far the students are traveling from?
- Could the location of a new school change who lives close enough to bike/walk versus who may need to take the bus?

5.4.4 Your Turn!

1. Suwannee County School District is studying changes that will need to be made to the transportation department when building new schools. To determine this, it is important to understand how students get to school. The school district has a population of $5{,}959$ students, so a random sample of approximately 5% of the students in each school was conducted. The results of the survey are shown in the bar chart.



a. Show how to predict the number of students in the district who take the bus to get to school.
To predict the number of students who take the bus to school, round the total number of students $(5{,}959)$ *to a friendlier number* $(6{,}000)$ *and then multiply by* 0.05 *or* 5% *which results in* 300 *students who were surveyed.*
From the graph we know that about 130 of the approximately 300 students surveyed take the bus to school.
The scale factor of total number of students in the district to the students sampled is $\frac{6{,}000}{300} = 20$. *So, we can predict that roughly* $20 \times 130 = 2{,}600$ *students in the district ride the bus to school.*

b. Show how to predict the number of students in the district who ride their bike to school.
To predict the number of students who ride their bike to school, we can use the information from part a. that roughly 300 students were surveyed, and the scale factor of students surveyed to the total number of students in the district is 20.
From the graph, we know $\frac{65}{300}$ *students surveyed ride their bike to school. So, we can predict that roughly* $20\ x\ 65 = 1{,}300$ *students in the district ride their bike to school.*

c. Whether or not a student can take the bus to school depends on how close they live to the school. Explain how this data set does not take this into consideration and what effect that might have on the district's decision.
Answers will vary. This data set does not include information about how far the students are traveling to get to school. There may be students who ride their bike who are actually within the zone for bus riding but prefer to get the exercise of riding their bikes. It may be useful for the district to conduct a follow up amongst students who are far enough away to be bussed to school to better assess their population size.

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5.4.5 Wrap-Up: Ticket Out the Door

Lin has one month before the seventh-grade class presidential election, and she wants to determine how likely she is to win. Since she can't ask all 665 seventh graders who they are voting for, she conducts a random sample of 105 seventh-grade students. The results of the survey are in the table.

Candidate	Number of People Planning to Vote for the Candidate
Lin	42
Beslinda (Lin's opponent)	30
Undecided	33

1. Use either percent, scale factor, or equivalent fraction to predict how many of the 665 total seventh-grade students currently plan on voting for Lin. Show your work.

Scale Factor	Percent	Equivalent Fraction
$\frac{665}{105} = 6\frac{1}{3}$ $6\frac{1}{3} \times 42 = 266$	$\frac{42}{105} = 40\%$ $0.4 \times 665 = 266$	$\frac{42}{105} = \frac{2}{5} \times \left(\frac{133}{133}\right)$ $= \frac{266}{665}$

2. Use another strategy to predict how many of the total seventh-grade voters are still undecided.

Scale Factor	Percent	Equivalent Fraction
$\frac{665}{105} = 6\frac{1}{3}$ $6\frac{1}{3} \times 33 = 209$	$\frac{33}{105} = 31.4286\%$ $0.314286 \times 665 = 209$	$\frac{33}{105} = \frac{11}{35} \times \left(\frac{19}{19}\right)$ $= \frac{209}{665}$

5.4.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

Encourage students to use creative methods to represent their reasoning using models or diagrams. For example, students could use a double number line or tape diagram to represent equivalent fractions. Take note of students who model using diagrams and consider pairing those students in the next lesson with partners who used only scale factor.

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